

Building Effective Machine Learning Solutions with **Open Source Tools**

Extracting hidden insights and patterns from data is of immense value to any business. Machine learning addresses this need effectively. But for it to work well, the right tools need to be used. Let's take a look at a few open source tools that could be put to good use in this domain.

The adoption of machine learning is not only due to the availability of more cost-effective and powerful data acquisition technologies and processing capabilities in the hardware domain. The proliferation of many tools, especially in the open source domain, makes machine learning easier to implement.

Along with the right set of enablers, machine learning is becoming pervasive in both enterprise as well as consumer applications. The usage of this technology can really be widespread, ranging from telematics to product recommendations, and from speech recognition to condition monitoring. No wonder, machine learning and the related technologies are rapidly becoming an integral part of the enterprise technology stack. Today it is considered as a critical skill of the developers as well as of the architects.

While machine learning can act as a key enabler to any business, for it to work effectively, it is crucial to choose the right set of tools. There are a wide range of tools available in the open source domain. Each one of them has evolved to cater to some specific need and has its own set of use cases. It is not fair to do an apple-to-apple comparison of all tools available today. Rather, it is necessary to analyse and understand the business need at hand and, accordingly, choose the tool that would be the best fit to solve the problem.

In this article, with the help of some relevant examples, we will discuss some of the key machine learning use cases along with the open source tool sets suitable for each of them.

Machine learning: A quick preview

Before delving into the machine learning tool set, let us take a quick look at what machine learning is all about and how it can be useful for business. Machine learning involves a system that acquires intelligence from past data, experience, observation and training. As part of machine learning, statistical techniques are used to learn the patterns, trends and structure hidden in the data. The learnings or the insights are then leveraged to perform many business operations. For example, by identifying books that have been purchased by your friends, Amazon is able to recommend books to you. Similarly, if you have a habit of watching the latest action



movies on Netflix, it will suggest other action movies for you, by using your past viewing history.

Open source tools

Till recently, machine learning was confined more to the world of academics. That may be the reason for the plethora of open source tools and libraries around machine learning and related technologies. With the passage of time and the vast interest in this area, many open source tools and libraries have evolved. By simply using the most suitable implementation of machine learning algorithms, it is possible to derive insights fairly quickly, easily and in a cost-effective manner. Some of the popular tools are listed below.

- **R:** This is a language and environment for statistical computing and graphics. R provides a wide variety of statistical (linear and non-linear regression, classical statistical tests, time-series analysis, classification, clustering, etc) and graphical techniques, and is highly extensible. The R language is often the vehicle of choice for research in statistical methodology, and it provides an open source route to participation in that activity.